

# Datos Multivariados

## Estadística Computacional

Juan Zamora Osorio  
juan.zamora@pucv.cl

Instituto de Estadística  
Pontificia Universidad Católica de Valparaíso

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PONTIFICIA  
UNIVERSIDAD  
CATÓLICA DE  
VALPARAÍSO

## Datos bivariados

| id Pasajero | Clase | Satisfacción |
|-------------|-------|--------------|
| 1           | E     | 2            |
| 2           | E     | 4            |
| 3           | E     | 1            |
| 4           | B     | 3            |
| 5           | E     | 1            |
| 6           | B     | 2            |
| 7           | P     | 4            |
| 8           | E     | 3            |
| 9           | E     | 2            |
| 10          | B     | 4            |
| 11          | E     | 3            |
| 12          | B     | 3            |
| ⋮           | ⋮     | ⋮            |

## Datos bivariados categóricos – tabla de contingencia

| id Pasajero | Clase | Satisfacción |
|-------------|-------|--------------|
| 1           | E     | 2            |
| 2           | E     | 4            |
| 3           | E     | 1            |
| 4           | B     | 3            |
| 5           | E     | 1            |
| 6           | B     | 2            |
| 7           | P     | 4            |
| 8           | E     | 3            |
| 9           | E     | 2            |
| 10          | B     | 4            |
| 11          | E     | 3            |
| 12          | B     | 3            |
| ⋮           | ⋮     | ⋮            |

|       |   | Satisfacción |   |   |   | Total |
|-------|---|--------------|---|---|---|-------|
|       |   | 1            | 2 | 3 | 4 |       |
| Clase | E | 2            | 2 | 2 | 1 | 7     |
|       | B | 0            | 1 | 2 | 1 | 4     |
|       | P | 0            | 0 | 0 | 1 | 1     |
| Total |   | 2            | 3 | 4 | 3 | 12    |

## Datos bivariados categóricos – tabla de contingencia

|       |   | Satisfacción |    |    |    | Total |
|-------|---|--------------|----|----|----|-------|
|       |   | 1            | 2  | 3  | 4  |       |
| Clase | E | 10           | 33 | 15 | 4  | 62    |
|       | B | 0            | 3  | 20 | 2  | 25    |
|       | P | 0            | 0  | 5  | 8  | 13    |
| Total |   | 10           | 36 | 40 | 14 | 100   |

### Clases bivariadas

- ▶ (E, 1), (E, 2), (E, 3), (E, 4), (B, 2), (B, 3), (B, 4), (P, 3), (P, 4).
- ▶ ¡Pueden ser muchas!

## Datos bivariados categóricos – tabla de contingencia

|       |          | Y             |          |               |          |               | Total        |
|-------|----------|---------------|----------|---------------|----------|---------------|--------------|
|       |          | $y_1$         | $\cdots$ | $y_j$         | $\cdots$ | $y_J$         |              |
| X     | $x_1$    | $n_{11}$      | $\cdots$ | $n_{1j}$      | $\cdots$ | $n_{1J}$      | $n_{1\cdot}$ |
|       | $\vdots$ | $\vdots$      | $\ddots$ | $\vdots$      | $\ddots$ | $\vdots$      | $\vdots$     |
|       | $x_k$    | $n_{k1}$      | $\cdots$ | $n_{kj}$      | $\cdots$ | $n_{kJ}$      | $n_{k\cdot}$ |
|       | $\vdots$ | $\vdots$      | $\ddots$ | $\vdots$      | $\ddots$ | $\vdots$      | $\vdots$     |
|       | $x_K$    | $n_{K1}$      | $\cdots$ | $n_{Kj}$      | $\cdots$ | $n_{KJ}$      | $n_{K\cdot}$ |
| Total |          | $n_{\cdot 1}$ | $\cdots$ | $n_{\cdot j}$ | $\cdots$ | $n_{\cdot J}$ | $n$          |

### Frecuencias absolutas marginales

$$n = \sum_{k=1}^K n_{k\cdot} = \sum_{j=1}^J n_{\cdot j} = \sum_{k=1}^K \sum_{j=1}^J n_{kj}.$$

## Datos bivariados categóricos – tabla de contingencia

|       |          | Y             |          |               |          |               | Total        |
|-------|----------|---------------|----------|---------------|----------|---------------|--------------|
|       |          | $y_1$         | $\cdots$ | $y_j$         | $\cdots$ | $y_J$         |              |
| X     | $x_1$    | $f_{11}$      | $\cdots$ | $f_{1j}$      | $\cdots$ | $f_{1J}$      | $f_{1\cdot}$ |
|       | $\vdots$ | $\vdots$      | $\ddots$ | $\vdots$      | $\ddots$ | $\vdots$      | $\vdots$     |
|       | $x_k$    | $f_{k1}$      | $\cdots$ | $f_{kj}$      | $\cdots$ | $f_{kJ}$      | $f_{k\cdot}$ |
|       | $\vdots$ | $\vdots$      | $\ddots$ | $\vdots$      | $\ddots$ | $\vdots$      | $\vdots$     |
|       | $x_K$    | $f_{K1}$      | $\cdots$ | $f_{Kj}$      | $\cdots$ | $f_{KJ}$      | $f_{K\cdot}$ |
| Total |          | $f_{\cdot 1}$ | $\cdots$ | $f_{\cdot j}$ | $\cdots$ | $f_{\cdot J}$ | 1            |

### Frecuencias relativas marginales

$$1 = \sum_{k=1}^K f_{k\cdot} = \sum_{j=1}^J f_{\cdot j} = \sum_{k=1}^K \sum_{j=1}^J f_{kj}.$$

## Datos bivariados categóricos – frecuencias condicionales

|       |          | Y             |          |               |          |               | Total        |
|-------|----------|---------------|----------|---------------|----------|---------------|--------------|
|       |          | $y_1$         | $\cdots$ | $y_j$         | $\cdots$ | $y_J$         |              |
| X     | $x_1$    | $f_{11}$      | $\cdots$ | $f_{1j}$      | $\cdots$ | $f_{1J}$      | $f_{1\cdot}$ |
|       | $\vdots$ | $\vdots$      | $\ddots$ | $\vdots$      | $\ddots$ | $\vdots$      | $\vdots$     |
|       | $x_k$    | $f_{k1}$      | $\cdots$ | $f_{kj}$      | $\cdots$ | $f_{kJ}$      | $f_{k\cdot}$ |
|       | $\vdots$ | $\vdots$      | $\ddots$ | $\vdots$      | $\ddots$ | $\vdots$      | $\vdots$     |
|       | $x_K$    | $f_{K1}$      | $\cdots$ | $f_{Kj}$      | $\cdots$ | $f_{KJ}$      | $f_{K\cdot}$ |
| Total |          | $f_{\cdot 1}$ | $\cdots$ | $f_{\cdot j}$ | $\cdots$ | $f_{\cdot J}$ | 1            |

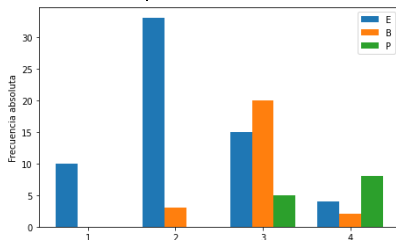
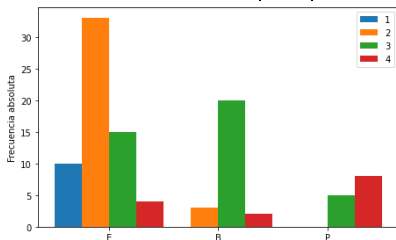
### Frecuencias condicionales

$$f_{k|j} = \frac{f_{kj}}{f_{\cdot j}} = \frac{n_{kj}}{n_{\cdot j}},$$

$$f_{j|k} = \frac{f_{kj}}{f_{k\cdot}} = \frac{n_{kj}}{n_{k\cdot}}.$$

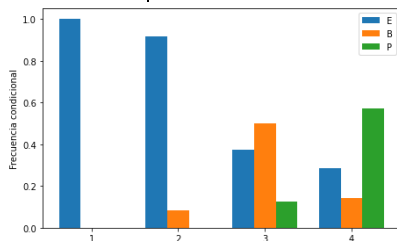
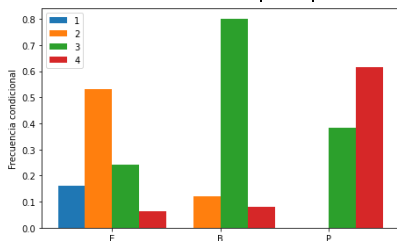
## Ejemplo – satisfacción de vuelos

|       |   | Satisfacción |    |    |    | Total |
|-------|---|--------------|----|----|----|-------|
|       |   | 1            | 2  | 3  | 4  |       |
| Clase | E | 10           | 33 | 15 | 4  | 62    |
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| Total |   | 10           | 36 | 40 | 14 | 100   |



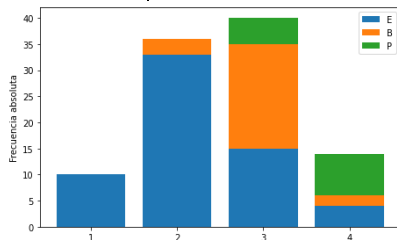
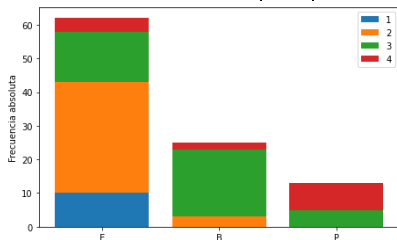
## Ejemplo – satisfacción de vuelos

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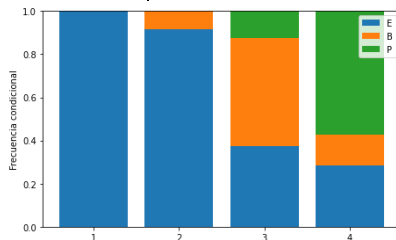
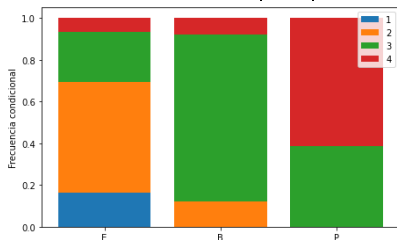
## Ejemplo – satisfacción de vuelos

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## Ejemplo – satisfacción de vuelos

|       |   | Satisfacción |    |    |    | Total |
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| Total |   | 10           | 36 | 40 | 14 | 100   |



# Independencia

|       |   | Satisfacción |    |    |    | Total |
|-------|---|--------------|----|----|----|-------|
|       |   | 1            | 2  | 3  | 4  |       |
| Clase | E | 10           | 33 | 15 | 4  | 62    |
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Se busca

►  $f_{k|j} = f_{k\cdot}$  y  $f_{j|k} = f_{\cdot j}$ .

Luego

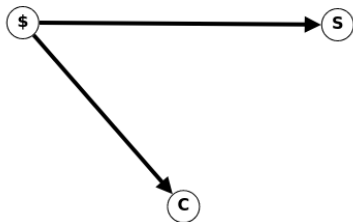
$$f_{kj} = f_{k\cdot} \cdot f_{\cdot j}.$$

► Esperamos frecuencia absoluta

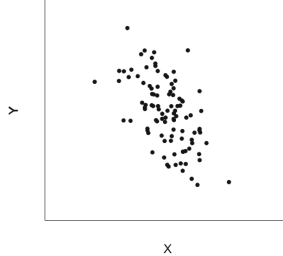
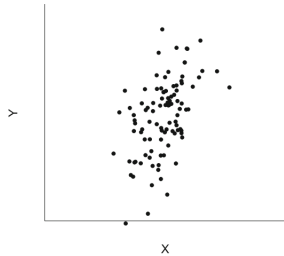
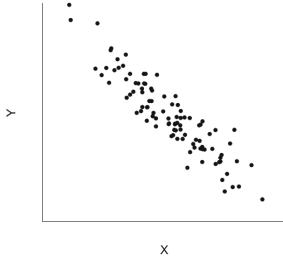
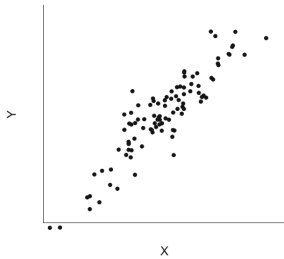
$$n_{kj} = n f_{kj} = n f_{k\cdot} \cdot f_{\cdot j} = n \frac{n_{k\cdot}}{n} \cdot \frac{n_{\cdot j}}{n} = \frac{n_{k\cdot} \cdot n_{\cdot j}}{n}.$$

# Independencia

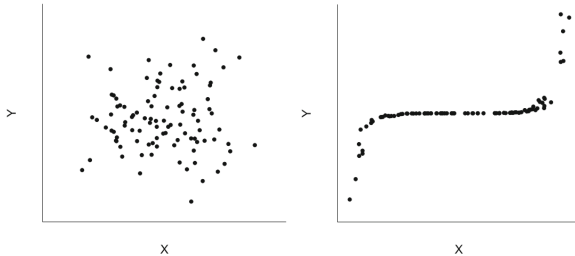
|       |   | Satisfacción |    |    |    | Total |
|-------|---|--------------|----|----|----|-------|
|       |   | 1            | 2  | 3  | 4  |       |
| Clase | E | 10           | 33 | 15 | 4  | 62    |
|       | B | 0            | 3  | 20 | 2  | 25    |
|       | P | 0            | 0  | 5  | 8  | 13    |
| Total |   | 10           | 36 | 40 | 14 | 100   |



# Datos bivariados continuos

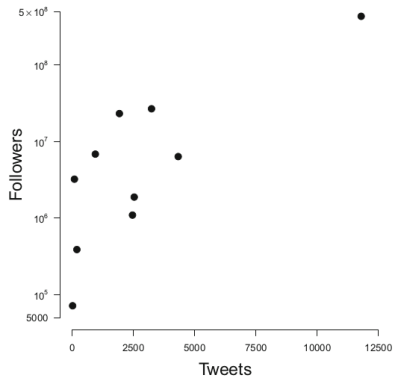


# Gráficos de dispersión (*scatter*)

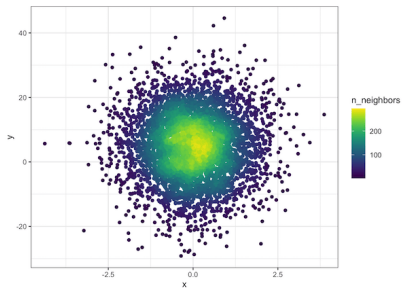
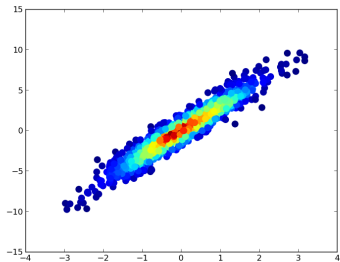


# Gráficos de dispersión (*scatter*)

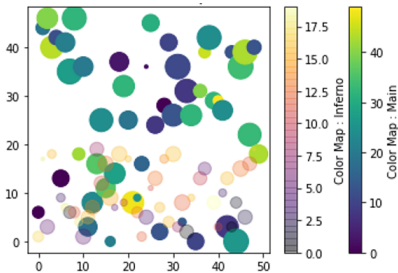
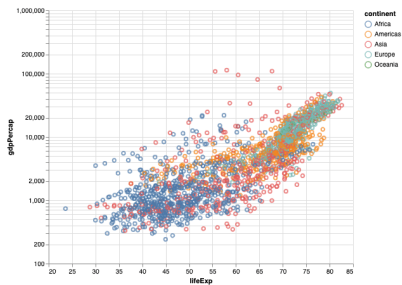
| Name              | Tweets | Followers  |
|-------------------|--------|------------|
| Angela Merkel     | 25     | 7194       |
| Barack Obama      | 11,800 | 43,400,000 |
| Jacob Zuma        | 99     | 324,000    |
| Dilma Rousseff    | 1934   | 2,330,000  |
| Sauli Niinistö    | 199    | 39,000     |
| Vladimir Putin    | 2539   | 189,000    |
| Francois Hollande | 4334   | 639,000    |
| David Cameron     | 952    | 688,000    |
| Enrique P. Nieto  | 3245   | 2,690,000  |
| John Key          | 2468   | 110,000    |



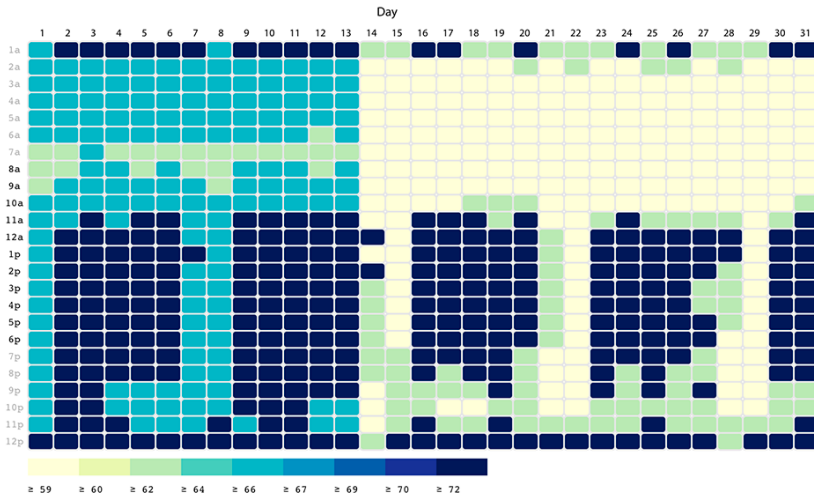
# Gráficos de dispersión (*scatter*)



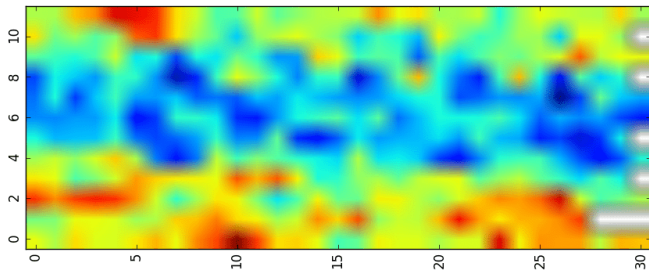
# Gráficos de dispersión (*scatter*)



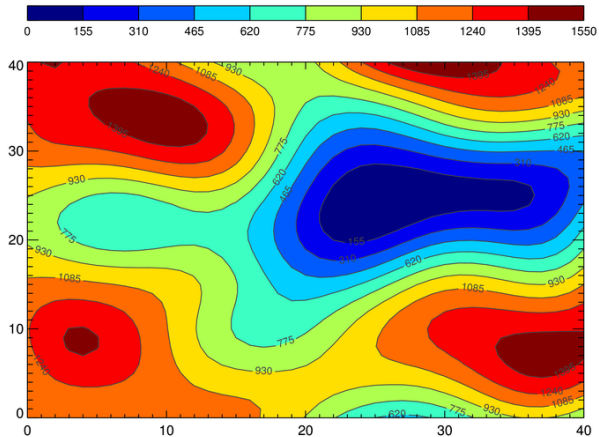
# Mapas de calor



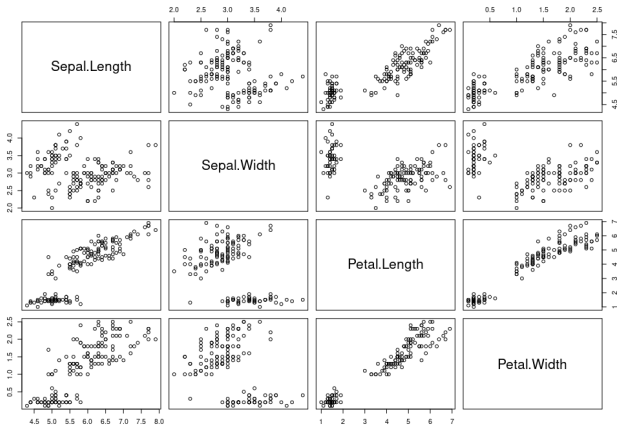
# Mapas de calor



# Mapas de calor – contorno



# Datos multivariados



# Correlación

## Covarianza

$$\text{Cov}(x, y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}).$$

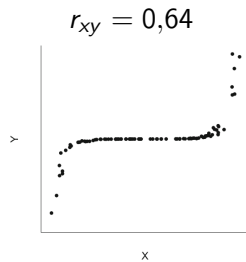
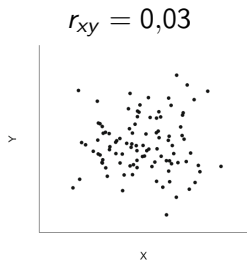
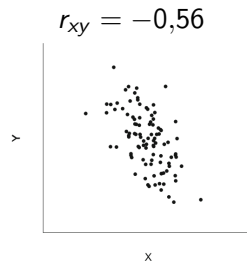
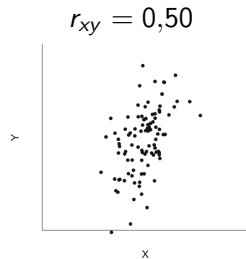
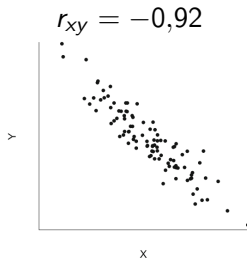
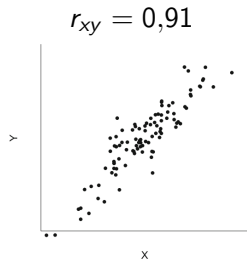
- ▶ Indica dependencia *lineal*.
- ▶ Notar  $\text{Cov}(x, x) = s_x^2$ .

## Correlación

$$r_{xy} = \frac{\text{Cov}(x, y)}{s_x s_y}.$$

- ▶ Indica dependencia *lineal*.
- ▶ Se puede mostrar que  $-1 \leq r_{xy} \leq 1$ .

# Correlación



# Matriz de varianzas y covarianzas

$$\mathbf{S} = \begin{bmatrix} \text{Cov}(x^1, x^1) & \cdots & \text{Cov}(x^1, x^j) & \cdots & \text{Cov}(x^1, x^J) \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \text{Cov}(x^j, x^1) & \cdots & \text{Cov}(x^j, x^j) & \cdots & \text{Cov}(x^j, x^J) \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \text{Cov}(x^J, x^1) & \cdots & \text{Cov}(x^J, x^j) & \cdots & \text{Cov}(x^J, x^J) \end{bmatrix}$$

## Propiedades

- ▶ Diagonal contiene las varianzas.
- ▶ Simétrica:  $\mathbf{S} = \mathbf{S}^T$ .
- ▶ Semidefinida positiva:
  - ▶  $\forall a \in \mathbb{R}^J, a^T \mathbf{S} a \geq 0$ .
  - ▶ Valores propios son todos no negativos.

# Matriz de varianzas y covarianzas

## Matriz de datos

$$\mathbf{X} = \begin{bmatrix} x_1^1 & \cdots & x_1^j & \cdots & x_1^J \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ x_i^1 & \cdots & x_i^j & \cdots & x_i^J \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ x_n^1 & \cdots & x_n^j & \cdots & x_n^J \end{bmatrix} = \begin{bmatrix} \mathbf{x}_1^T \\ \vdots \\ \mathbf{x}_i^T \\ \vdots \\ \mathbf{x}_n^T \end{bmatrix}$$

## Construcción

$$\bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i,$$

$$\mathbf{S} = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T.$$

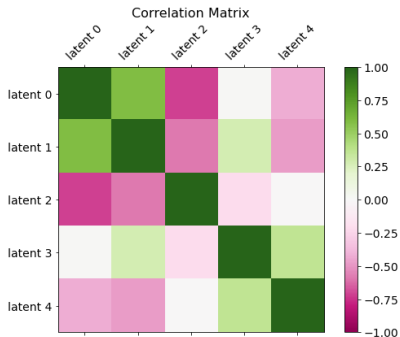
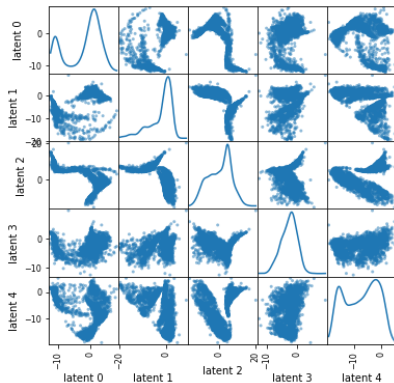
# Matriz de correlación

$$\mathbf{R} = \begin{bmatrix} 1 & \cdots & r_{1j} & \cdots & r_{1J} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ r_{j1} & \cdots & 1 & \cdots & r_{jJ} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ r_{J1} & \cdots & r_{Jj} & \cdots & 1 \end{bmatrix}$$

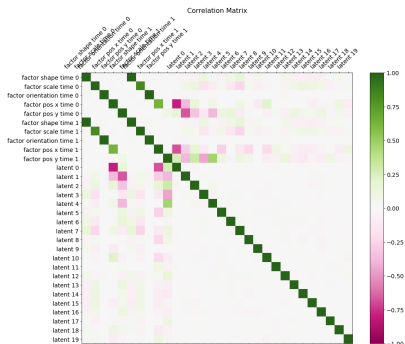
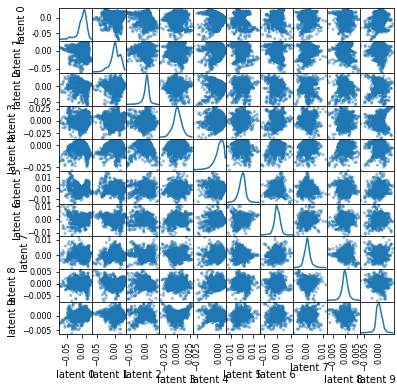
## Propiedades

- ▶ No es igual a la matriz de varianzas y covarianzas escalada.
- ▶ Se puede construir como **S** estandarizando los atributos primero.

# Matriz de correlación



# Matriz de correlación



# Matrices de correlación y de varianzas y covarianzas

